SET A- Q1. Use privacy and security settings on operating system.

Password Setting:

1. Click on Start Menu → Settings
2. Open Privacy & Security
3. Go to Accounts → Sign-in options
4. Click on Password → Add/Change
5. Set a strong password

Windows Security (Antivirus):

1. Go to Privacy & Security → Windows Security
2. Click Virus & Threat Protection
3. Turn ON:
    - Real-time protection
    - Cloud-delivered protection

Firewall Settings

1. Open Control Panel → System and Security
2. Click Windows Defender Firewall
3. Click Turn Windows Defender Firewall ON/OFF
4. Enable firewall for both Private and Public networks

Privacy Settings (Application Permissions):

Location:
1. Go to **Privacy & Security → Location**
2. Turn Location ON/OFF as required

Camera:
1. Go to **Privacy & Security → Camera**
2. Enable/Disable camera access for apps

Microphone:
1. Go to **Privacy & Security → Microphone**
2. Enable/Disable microphone access

Windows Update:

1.Go to Settings → Windows Update
2.Click Check for updates and install updates

Screen Lock:
1. Go to Settings → Personalization → Lock Screen
2. Set screen timeout and enable auto lock

Result: Privacy and security settings were successfully configured on the system.

Q2.Write steps to change password of your local account.

→ Procedure:→

1. Click on Start Menu → Settings
2. Select Accounts
3. Click on Sign-in options
4. Under Password, click on Change
5. Enter your current password and click Next
6. Enter a new password
7. Re-enter the new password to confirm
8. Add a password hint
9. Click on Next → Finish

Set B- Q1. Write a utility using C/Shell programming to create strong password authentication (Password should be more than 8characters, and combination of digits, letters and special characters #, %, &, @)

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main() {
    char password[100];
    int i, hasDigit=0, hasLetter=0, hasSpecial=0;
    printf("Enter Password: ");
    scanf("%s", password);

    if(strlen(password) < 8) {
        printf("Password must be at least 8 characters long\n");
        return 0;
    }

    for(i = 0; password[i] != '\0'; i++) {
        if(isdigit(password[i]))
            hasDigit = 1;
        else if(isalpha(password[i]))
            hasLetter = 1;
        else if(password[i]=='#' || password[i]=='%' || password[i]=='&' ||
password[i]=='@')
            hasSpecial = 1;
    }

    if(hasDigit && hasLetter && hasSpecial)
        printf("Strong Password Accepted\n");
    else
        printf("Password must contain digit, letter and special character (#, %%, &,
@)\n");
    return 0;
}
```

Output:→

Weak Password

Enter Password: abc12
Password must be at least 8 characters long

Invalid Combination

Enter Password: abcdefgh
Password must contain digit, letter and special character (#, %, &, @)

Strong Password

Enter Password: Abc123@#
Strong Password Accepted

Q2.List various Password attacks and explain any one.
→

List of Password Attacks:

1. Brute Force Attack,
2. Dictionary Attack,
3. Phishing Attack,
4. Keylogger Attack,
5. Shoulder Surfing
6. Social Engineering Attack,

Explanation of Brute Force Attack:→

Brute Force Attack is a method in which an attacker tries **all possible combinations of passwords** until the correct password is found.

- It uses automated tools
- It can take a long time if password is strong
- Weak passwords can be cracked quickly

Example:

If password is 1234, attacker will try:
0000 → 0001 → 0002 → … → 1234

SET C → Q1. Write a C program to implement caesar cipher technique to perform encryption and decryption of text.

```
#include <stdio.h>
#include <ctype.h>
void encrypt(char text[], int key) {
    int i;
    for(i = 0; text[i] != '\0'; i++) {
        char ch = text[i];

        if(isalpha(ch)) {
            char base = islower(ch) ? 'a' : 'A';
            text[i] = (ch - base + key) % 26 + base;
        }
    }
}

void decrypt(char text[], int key) {
    int i;
    for(i = 0; text[i] != '\0'; i++) {
        char ch = text[i];

        if(isalpha(ch)) {
            char base = islower(ch) ? 'a' : 'A';
            text[i] = (ch - base - key + 26) % 26 + base;
        }
    }
}

int main() {
    char text[100];
    int key;

    printf("Enter text: ");
    scanf("%s", text);

    printf("Enter key: ");
    scanf("%d", &key);

    encrypt(text, key);
    printf("Encrypted text: %s\n", text);

    decrypt(text, key);
    printf("Decrypted text: %s\n", text);

    return 0;
}
```

Output:

Enter text: HELLO
Enter key: 3
Encrypted text: KHOOR
Decrypted text: HELLO

Q2. Consider plaintext and convert into ciphertext using Caesar Cipher. (Shift= 3)

(i) computer security

c → f
o → r
m → p
p → s
u → x
t → w
e → h
r → u

(space)

s → v
e → h
c → f
u → x
r → u
i → l
t → w
y → b
N → Q
e → h
t → w
w → z
o → r
r → u
k → n

Ciphertext:

frpsxwhu vhfxulwb

(ii) Network information

N → Q
e → h
t → w
w → z
o → r
r → u
k → n

(space)

i → l
n → q
f → i
o → r
r → u
m → p
a → d
t → w
i → l
o → r
n → q

Ciphertext:

Qhwzrun lqirupdwlrq

SET D→ Q1. Implement rail fence encryption technique to perform encryption of text using C programming language.

```
#include <stdio.h>
#include <string.h>
int main() {
    char text[100];
    int key, i, j, k = 0;

    printf("Enter text: ");
    scanf("%s", text);

    printf("Enter number of rails (key): ");
    scanf("%d", &key);

    int len = strlen(text);
    char rail[key][len];

    // Fill matrix with '\n'
    for(i = 0; i < key; i++)
        for(j = 0; j < len; j++)
            rail[i][j] = '\n';

    int row = 0, dir = 1;

    // Fill characters in zig-zag manner
    for(i = 0; i < len; i++) {
        rail[row][i] = text[i];

        if(row == 0)
            dir = 1;
        else if(row == key - 1)
            dir = -1;

        row = row + dir;
    }

    // Read matrix row-wise to get cipher
    printf("Encrypted text: ");
    for(i = 0; i < key; i++) {
        for(j = 0; j < len; j++) {
            if(rail[i][j] != '\n')
                printf("%c", rail[i][j]);
        }
    }
    return 0;
}
```

Output:

Enter text: HELLOWORLD
Enter number of rails (key): 3
Encrypted text: HOLELWRDLO

Q2. Consider plaintext "Welcome to Computer World", encrypt with the help of Rail Fence technique (Rails= 3)

W				O				O				P				R				L	
	E		C		M		T		C		M		U		E		W		R		D
		L				E				O				T				O			

Ciphertext: WOOPRL ECMTCMUEWRD LEOTO

SET E→ Q1.Write a C program to implement Diffie-Hellman key exchange algorithm to perform encryption of text.

```
#include <stdio.h>
#include <math.h>

// function for power (a^b % mod)
long long power(long long a, long long b, long long mod) {
    long long result = 1;
    while(b > 0) {
        if(b % 2 == 1)
            result = (result * a) % mod;
        a = (a * a) % mod;
        b = b / 2;
    }
    return result;
}

int main() {
    long long p, g, a, b;
    long long A, B, keyA, keyB;

    // Step 1: Public values
    printf("Enter prime number (p): ");
    scanf("%lld", &p);
    printf("Enter primitive root (g): ");
    scanf("%lld", &g);

    // Step 2: Private keys
    printf("Enter private key of A: ");
    scanf("%lld", &a);
```

```
    printf("Enter private key of B: ");
    scanf("%lld", &b);

    // Step 3: Public keys
    A = power(g, a, p);
    B = power(g, b, p);
    printf("Public key of A: %lld\n", A);
    printf("Public key of B: %lld\n", B);

    // Step 4: Secret keys
    keyA = power(B, a, p);
    keyB = power(A, b, p);
    printf("Secret key at A: %lld\n", keyA);
    printf("Secret key at B: %lld\n", keyB);

    // Simple encryption using key
    char text[100];
    printf("Enter text: ");
    scanf("%s", text);
    printf("Encrypted text: ");
    for(int i = 0; text[i] != '\0'; i++) {
        printf("%c", text[i] + keyA);  // simple shift
    }
    return 0;
}
```

Output:

```
Enter prime number (p): 23
Enter primitive root (g): 5
Enter private key of A: 6
Enter private key of B: 15
```

Public key of A: 8
Public key of B: 19

Secret key at A: 2
Secret key at B: 2

Enter text: HELLO
Encrypted text: JGNNQ

Q2. Write the applications and limitations of Diffie-Hellman algorithm.
→

Applications: →

1. Secure Key Exchange: Used to securely share secret keys over an insecure network.

2. VPN (Virtual Private Network): Helps in creating secure tunnels for data transmission.

3. Cryptographic Protocols: Used as a base in many security algorithms and systems.

Limitations: →

1. No Authentication: Does not verify identity → vulnerable to Man-in-the-Middle attack.

2. Not an Encryption Algorithm: Only used for key exchange, not for actual data encryption.

3. Computationally Intensive: Requires large calculations → slower for low-power devices.

SET F → Q1. Configure firewall settings on any operating system.

Turn Firewall ON/OFF:→

1. Click on **Start Menu → Control Panel**
2. Select **System and Security**
3. Click on **Windows Defender Firewall**

4. Click on **Turn Windows Defender Firewall ON or OFF**
5. Select:
   - **Turn ON Firewall** for Private Network
   - **Turn ON Firewall** for Public Network
6. Click **OK**

Allow or Block an App through Firewall:→

1. Go back to **Windows Defender Firewall**
2. Click on **Allow an app or feature through Windows Defender Firewall**
3. Click on **Change Settings**
4. Check or uncheck apps to **Allow/Block access**
5. Click **OK**

Advanced Firewall Settings:→

1. Click on **Advanced Settings**
2. Select:
3. **Inbound Rules → New Rule**
4. Choose rule type (Port/Program)
5. Select:
6. Allow connection / Block connection
7. Click **Next → Finish**

Q2. Explain the needs of firewalls?

→Needs of Firewalls:

1.Network Security: It monitors incoming and outgoing traffic and allows only safe data.

2.Prevention of Cyber Attacks: Helps in preventing attacks like hacking, malware, and viruses.

3.Control Over Network Traffic: Allows or blocks specific applications, ports, or IP addresses.

4.Data Protection: Prevents sensitive data from being leaked outside the network.

5.Access Control: Restricts access to certain services or resources within a network.